**CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**
**CHANGA**
**VI Semester of BSc (Biological Sciences) Examination April-May 2019**

**MI 601 Fermentation Technology**

**Date: 27/04/2019 Day: Saturday Time: 1:30 p.m. to 3:30 p.m. Maximum Marks: 35**

**MCQ**

**Instructions:**
1) Provide all necessary information as required in the answer sheet.
2) Attempt all questions.
3) Questions in **Part A** should be answered in the question paper itself.
4) Use of cell phones is strictly prohibited.
5) Use of non-programmable calculators may be allowed if needed.

**PART A Time: 1:30 p.m. to 1:50 p.m. Marks: 10**

**Q. 1 Choose the correct answer from the options given below each question:**

1) Glutamic acid production is greatest when the biotin is present in the fermentation broth at growth limiting concentration by not allowing the following
(i) growth (ii) feedback inhibition (iii) floculation (iv) none of the above

2) $Mn^{+2}$ or $Mg^{+2}$ should be in limiting concentration during citric acid fermentation as they activate the following enzyme of TCA
(i) oxalosuccinic decarboxylase (ii) succinic dehydrogenase
(iii) isocitric dehydrogenase (iv) aconitase

P.T.O.

3) α amylases can attack the interior of starch to liberate the following

(i) glucose (ii) maltose (iii) triose (iv) all of the above

4) The following factor is required for the induction and derepression of enzymes necessary for alkaloid synthesis

(i) tryptophan (ii) mannitol (iii) lysine (iv) fumaric acid

5) Addition of phenylacetic acid to the fermentation broth leads to the synthesis of the following

(i) Penicillin G (ii) Penicillin V ((iii) Penicillin K (iv) Penicillin O

6) The growth rate of the following group of microorganisms is a rate limiting factor for biogas production

(i) Hydrolytic Bacteria (ii) Fermentative Acidogenic Bacteria

(iii) Acidogenic Bacteria (iv) methanogens

7) Oleaginous Microorganisms have the ability to amass lipids more than the following of their dry cell weight

(i) 20-40% (ii) 40-60% (iii) 10-30% (iv) 30-50 %

8) Polymers of the following are used as biodegradable thermoplastics

(i) citric acid (ii) lactic acid (iii) glutamic acid (iv) lignocellulose

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9) Carrier based Rhizobium biofertilizers should contain a minimum of the following CFU/gram

(i) $5 x 10^7$ (ii) $5 x 10^4$ (iii) $1 x 10^7$ (iv) $1 x 10^4$

10) High aeration rate during ethanol fermentation supports the production of the following

(i) ethanol (ii) biomass (iii) lactic acid (iv) acetone

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**PART B** **Time: 1:50 a.m. to 3:30 p.m.** **Marks: 25**

**SECTION -I**

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|---|---|---|
| **Q - II** | **Answer the following questions as directed** | **10** |
| **A** | **Differentiate between any one the following** | **(02)** |
| **i.** | Primary and secondary metabolites giving examples | |
| **ii.** | Semi synthesis and Biotransformation | |
| **B** | **Discuss briefly any three of the following** | **(06)** |
| **i.** | Fermentation Conditions Used in Steroid Transformation | |
| **ii.** | Therapeutic Uses of alkaloids | |
| **iii.** | Limitations of SCO production | |
| **iv.** | The desirable properties of an organisms to be used for biological control | |
| **C** | **Justify** Phenyl acetic acid should be added continuously at low concentration during penicillin fermentation | **(02)** |

**SECTION - II**

| | | |
|---|---|---|
| **Q-III** | **Answer the following questions as directed** | **15** |
| **A** | **Discuss briefly any two the following** | **(04)** |
| **i.** | Factors affecting ethanol recovery during ethanol fermentation | |
| **ii.** | Strain improvement strategies for amino acid production with example | |
| **iii.** | Downstream processing of lactic acid | |
| **B** | **Discuss briefly any one the following** | **(03)** |
| **i.** | Microbial consortium involved in biogas production | |
| **ii.** | Discuss briefly: the over production of citric acid by inhibiting the free-flow of TCA | |
| **C** | **Discuss briefly any four of the following** | **(08)** |
| **i.** | Application of enzymes in production of high fructose corn syrup | |
| **ii.** | Fermentative production of streptomycin | |
| **iii.** | Modification in medium composition to improve permeability of cells for glutamic acid fermentation | |
| **iv.** | Production of semisynthetic penicillin | |
| **v.** | Microbiological methods for the production of amino acid with examples | |